ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 1 SET C **WINTER SEMESTER, 2024-2025**

DURATION: 30 MINUTES **FULL MARKS: 20**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. The *UpSiDe DoWn* is a *strange* world where every vector from the real world appears shifted, rotated or rearranged in unexpected ways. Mr.Harrington and Mr.Henderson, two curious explorers studying this strange realm have discovered pairs of vectors that correspond between the two worlds. They observed $\mathbf{v_1} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v_2} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$ in the real world correspond to $\mathbf{u_1} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ and $\mathbf{u_2} = \begin{bmatrix} -4 \\ -6 \end{bmatrix}$ in the *UpSiDe DoWn* world respectively. And it is obvious that the standard basis vectors in the real world are $\hat{\mathbf{i}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$(for x-axis) and $\hat{\mathbf{j}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$(for y-axis). 3 + 2 (CO2) (PO1)

   (a) Determine the basis vectors of the *UpSiDe DoWn* world (i.e., the transformation matrix that converts real-world vectors into *UpSiDe DoWn* vectors).

   (b) Using your result from part (a), answer the following:

   (b)i. Objects in the *UpSiDe DoWn* world are: ☐ same ☐ enlarged ☐ squeezed

   (b)ii. The orientation in the *UpSiDe DoWn* world is: ☐ as usual ☐ flipped

   NB: Your answers for parts (b)(i) and (b)(ii) should be supported by brief mathematical justification. It is not necessary to show the steps of solving equations for this question.

2. You are provided with the following equations - 1 + 0.5 + 5 + 1 (CO2) (PO1)

$$3x - y + 2z = 3 \quad (1)$$

$$6x - 2y + 3z = 7 \quad (2)$$

$$x + 3y - z = 6 \quad (3)$$

(a) Write down the system of linear equations in $Ax = b$ or Matrix form.

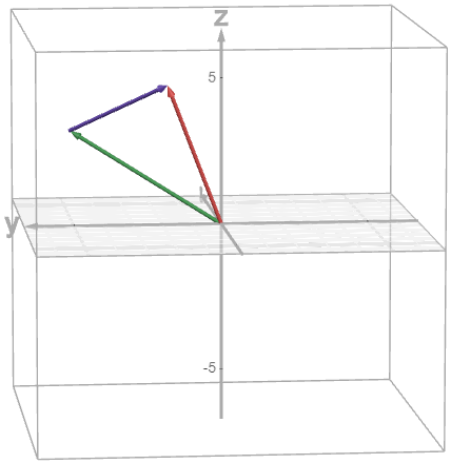
(b) Write down the augmented matrix ($[A|b]$).

(c) Perform necessary operations on the coefficient matrix $\mathbf{A}$ to obtain the upper triangular matrix $\mathbf{U}$ using elimination matrices $\mathbf{E}_{ij}$. Convert $\mathbf{A}x = \mathbf{b}$ to $\mathbf{U}x = \mathbf{c}$ (where $\mathbf{U}$ is an upper triangular matrix).

(d) Find out the values of x,y and z using back substitution.

3. Answer the following question by ticking the correct box (Question 3.a - Question 3.d) and writing the correct answer in the box (Question 3.c). 5 (CO2) (PO1)

a) What type of picture is shown below?



☐ Row Picture ☐ Column Picture

b) What is the span of two vectors $\vec{a}$ and $-\frac{2}{3}\vec{a}$?

☐ 3D space ☐ 2D space ☐ Line ☐ $(0,0)$

$$H = \begin{bmatrix} 152/19 & 24 & -32 \\ 117/13 & 27 & -36 \\ 46/23 & 6 & -8 \end{bmatrix}$$

c) What is the span of the columns of the matrix H?

☐ 3D space ☐ 2D space
☐ A line ☐ Origin

d) Which are the independent columns of the matrix H? (Multiple ticks are allowed)

☐ 1st ☐ 2nd ☐ 3rd ☐ None

e) What is the trace of the matrix H?

Answer:

4. Based on the following row operations **(applied in order)** to a matrix B of order 3×3, determine 2.5 (CO2) (PO1)

(a) The **elimination matrix** E that transforms B into the upper triangular matrix U.

(b) Its inverse, E^{-1} (i.e., the corresponding lower triangular matrix L).

(i) $r_2 = r_2 - 4 \times r_1$

(ii) $r_3 = r_3 - 1 \times r_2$

(iii) $r_3 = r_3 + 3 \times r_1$